Def. Let $(X, \mathcal{M})$ be a measurable space.
A map $\mu: \mathcal{M} \to \mathbb{R}$ (or $\mathbb{C}$) is said to be Countably additive if,
for any Pairwise disjoint Countable collection $\{A_i\}_{i \in \mathbb{N}}$,

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

A nonnegative-valued Countably additive μ is called a Positive measure,
a Complex-valued Countably additive μ is called a Complex measure.
The ordered triple $(X, \mathcal{M}, \mu)$ is called a measure space.

Thm. Let μ be a positive measure on $(X, \mathcal{M})$. Then,

a) $\mu(\emptyset) = 0$.

b) $\mu(A_1 \cup \cdots \cup A_n) = \mu(A_1) + \cdots + \mu(A_n)$ if $A_1, \cdots, A_n \in \mathcal{M}$ are pairwise disjoint.

c) For all $A, B \in \mathcal{M}$, $A \subseteq B$ implies $\mu(A) \leq \mu(B)$.

d) If $A_n \in \mathcal{M}$ $(n \in \mathbb{N})$ satisfies $A_k \subseteq A_{k+1}$ $(k \in \mathbb{N})$, then

$$\mu(A_n) \to \mu\left(\bigcup_{n \in \mathbb{N}} A_n\right) \text{ as } n \to \infty.$$

e) If $A_n \in \mathcal{M}$ $(n \in \mathbb{N})$ satisfy $A_k \supseteq A_{k+1}$ $(k \in \mathbb{N})$, and $\mu(A_1) < \infty$, then

$$\mu(A_n) \to \mu\left(\bigcap_{n \in \mathbb{N}} A_n\right) \text{ as } n \to \infty$$

Proof. Except c), the monotonicity, these hold for complex measure.

a). Take $A_1 = A \in \mathcal{M}$ so that $\mu(A) < \infty$, and $A_k = \emptyset$ for all $k \geq 2$. By the countably additivity,

$$\mu(A) = \mu(A) + \sum_{n=2}^{\infty} \mu(\emptyset) < \infty \quad \text{so that } \mu(\emptyset) = 0.$$

b). Take $A_k = \emptyset$ for all $k \geq n+1$, so proved.

c). Notice that $B = A \cup (B \setminus A)$ and $A \cap (B \setminus A) = \emptyset$.
By the finite additivity, $\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A)$.

d). Take $B_1 = A_1$ and put $B_n = A_n \setminus A_{n-1}$ for $n \geq 2$. Then, $B_n \in \mathcal{M}$ $(n \in \mathbb{N})$, and if $i > j$, then $B_i \cap B_j = (A_i \setminus A_{i-1}) \cap (A_j \setminus A_{j-1}) = \emptyset$, since $A_j \subseteq A_{i-1}$.
Moreover, $A_n = \bigcup_{i=1}^{n} B_i$, so that

$$\mu(A_n) = \sum_{i=1}^{n} \mu(B_i) \to \sum_{i=1}^{\infty} \mu(B_i) = \mu\left(\bigcup_{i=1}^{\infty} B_i\right) = \mu\left(\bigcup_{n \in \mathbb{N}} A_n\right)$$

e). Take $C_n = A_1 \setminus A_n$. Then, $C_1 \subseteq C_2 \subseteq \cdots$, and $C_n \cup A_n = A_1$, $C_n \cap A_n = \emptyset$, therefore $\mu(C_n) = \mu(A_1) - \mu(A_n)$ and

$$A_1 \setminus \left(\bigcap_{n \in \mathbb{N}} A_n\right) = A_1 \cap \left(\bigcup_{n \in \mathbb{N}} A_n^c\right) = \bigcup_{n \in \mathbb{N}} A_1 \cap A_n^c = \bigcup_{n \in \mathbb{N}} C_n,$$ so that by d),

$$\mu(A_1) - \mu\left(\bigcap_{n \in \mathbb{N}} A_n\right) = \mu\left(A_1 \setminus \left(\bigcap_{n \in \mathbb{N}} A_n\right)\right) = \mu(A_1) - \lim_{n \to \infty} \mu(A_n)$$ ☐